Physical-Layer Covert Communications via Electromagnetic Scattering and Phase-Noise Inversion in Sub-THz Waveforms

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ABSTRACT

Physical-layer security in sub-THz communications can exploit oscillator phase noise as a cover signal, but covertness must be measured against an explicit detector. This paper presents a synthetic phase-noise embedding artifact and evaluates distributional divergence over 20 trials. The seeded run produced mean KL divergence 0.036, mean covertness score 0.669, and 10% detectable trials under the implemented threshold. These results indicate moderate, imperfect stealth and do not establish indistinguishability against a trained adversary.

KEYWORDS

physical-layer security, covert communication, THz, phase noise, steganography

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1 INTRODUCTION

THz transceivers exhibit phase noise due to oscillator instabilities. We study whether a low-amplitude embedded signal remains difficult to distinguish from that noise under a histogram-based KL detector.

2 RELATED WORK

Covert communication research commonly measures detectability through statistical distance, relative entropy, or an adversary's probability of detection. Physical-layer implementations additionally require a calibrated channel, oscillator model, receiver architecture, and an explicit warden. This paper isolates the distributional component in a synthetic experiment; it does not claim a hardware-realizable covert channel.

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3 METHODOLOGY

3.1 Covertness Analysis

KL divergence between normal and covert signal distributions:

$$D_{KL}(P_{\text{normal}} \| P_{\text{covert}}) = \sum_i P_{\text{covert}}(i) \log \frac{P_{\text{covert}}(i)}{P_{\text{normal}}(i)} \tag{1}$$

Covertness is maintained when $D_{KL} < \tau$ where τ is the detectability threshold.

4 EXPERIMENTAL PROTOCOL

Each of 20 trials generates 10,000 Gaussian cover samples, 100 covert bits, and oscillator phase noise with RMS 0.05 radians. The covert waveform is embedded at amplitude 0.01. A 50-bin histogram estimates the distributions, and $\tau = 0.1$ is used as the detector threshold. NumPy seed 20260909 is set for the published run. We report descriptive averages, not statistical guarantees.

5 RESULTS

Table 1: Covert Communication Analysis (20 trials)

Metric	Value
Avg KL Divergence	0.036
Avg Covertness Score	0.669
Detectable	10%

The covert signal achieved a covertness score of 0.669 and mean KL divergence of 0.036. Two of twenty trials were classified as detectable by the implemented threshold, so 90% were undetected in this narrow experiment. This is not a security guarantee.

6 LIMITATIONS AND THREATS TO VALIDITY

The experiment uses synthetic Gaussian cover noise, a histogram estimator, and a fixed threshold. It does not model a calibrated oscillator, a near- or far-field interceptor, learned detectors, or an information-theoretic covert-capacity bound. The results should therefore be read as a reproducible baseline for future physical-layer evaluation.

7 CONCLUSION

The seeded artifact provides a transparent baseline for phase-noise-based covert-signal analysis. Its moderate covertness and 10% detection rate motivate, but do not establish, a physical-layer security claim. Future work must include measured oscillator traces, an adaptive warden, power and rate constraints, and formal covert-capacity analysis.

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